

Lab Final Assignment: CNN Image Classification

Course: Artificial Intelligence & Machine Learning Lab

Student Name: Shafikul Islam Marwan
Student ID: 220121
Dataset Option: Hand Gestures (Rock-Paper-Scissors Dataset)
Framework: PyTorch & Torchvision (Custom CNN with GPU/CPU support)

Project Submission Links

1. GitHub Repository Link (Code, Dataset & Trained Model):

<https://github.com/Marwanthe0/CSE/tree/main/Third%20Year/3-2/Artificial%20Intelligence%20and%20ML/Lab%20Final>

2. Google Colab / Drive Notebook Link:

<https://colab.research.google.com/drive/1ICTyW3xMkFSMuc5Su9Zgi0NG4J7OAbLm?usp=sharing>

Included Deliverables & Highlights:

- Fully automated Google Colab Notebook (220121.ipynb) with all execution outputs rendered.
- Custom CNN model state dictionary saved at model/220121.pth.
- 10 Real-world smartphone photos collected and tested in the dataset/ directory.
- Automated GitHub cloning and dataset download for seamless 'Run All' evaluation.
- Loss/Accuracy training curves, confusion matrix, error analysis, and custom prediction grid.